

Golf Putter Term Project

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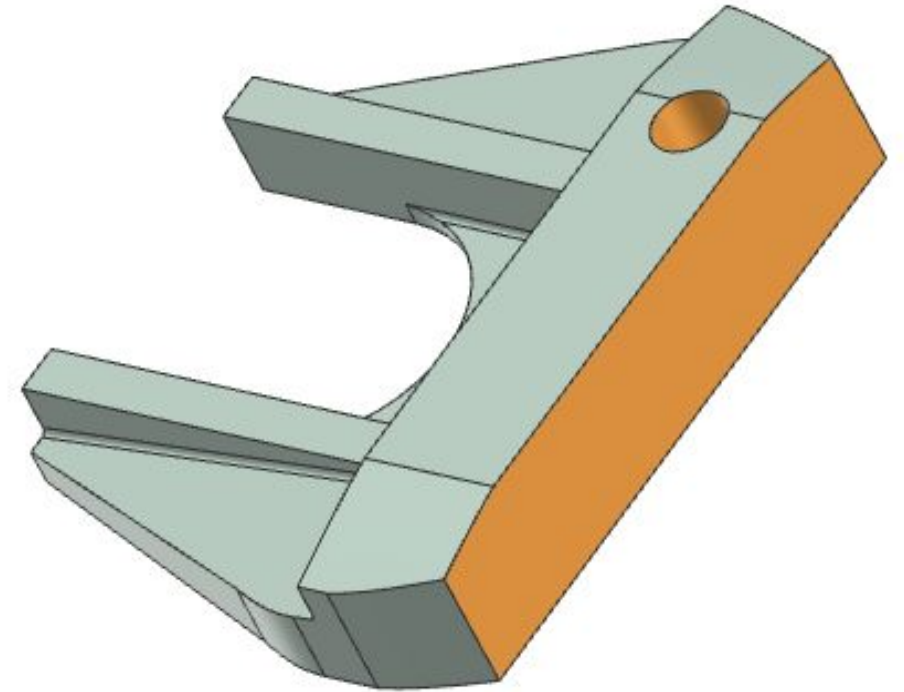
MFGE 413

Fall 2025



Product – Golf Putter Head

- Use – For Golf Putters
- Market – Hobbyist Golfers
- Value – Allows for players to have custom personalized equipment





Planned Output



Daily Output Goal

5 parts



Working Days per year

260 days



Annual Output

1300 Parts



Manufacturing Plan – Machine Choice

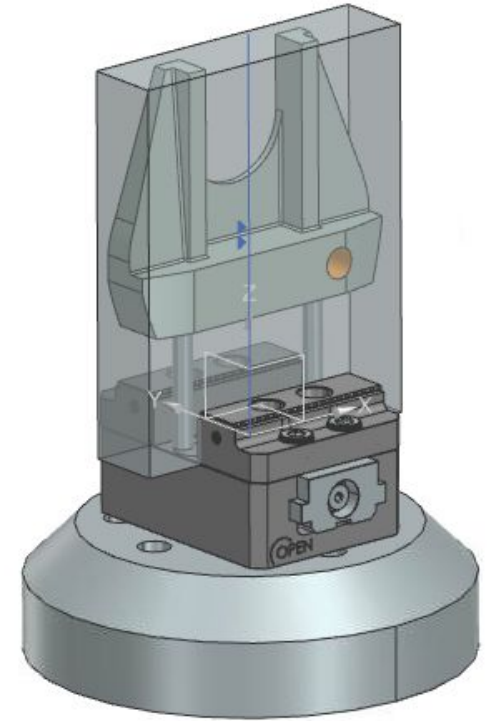
Machine	DMG MORI DMU 40 – 5AX
Size	21.7in x 17.7in x 15.5in
Spindle Speed	15000 rpm
Tool Changer	24 different tools
User Interface	SIEMENS
Cost	Starting \$250,000
Competitive Advantage	Palletization and Automatization Potential





Manufacturing Plan – Fixturing & Tooling

- 2513N1 Compact Milling–Machine Vise
- 2549N7 2 High Riser for Dovetail Milling–Machine Vise
- 2 Tubes connection to leftover blank in vise





Manufacturing Plan – Pre & Post Process

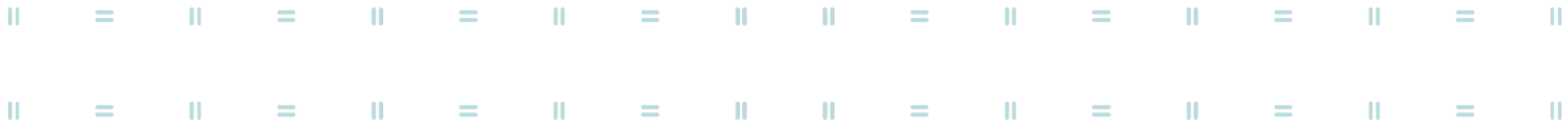
Pre-Process

1. Purchase large billets
2. Bandsaw down to Blank size
3. Insert into fixturing/tooling



Post-Process

1. Remove from Fixturing/Tooling
2. Mill off fixturing tubes
3. Deburr/Polish part
4. Inspecting

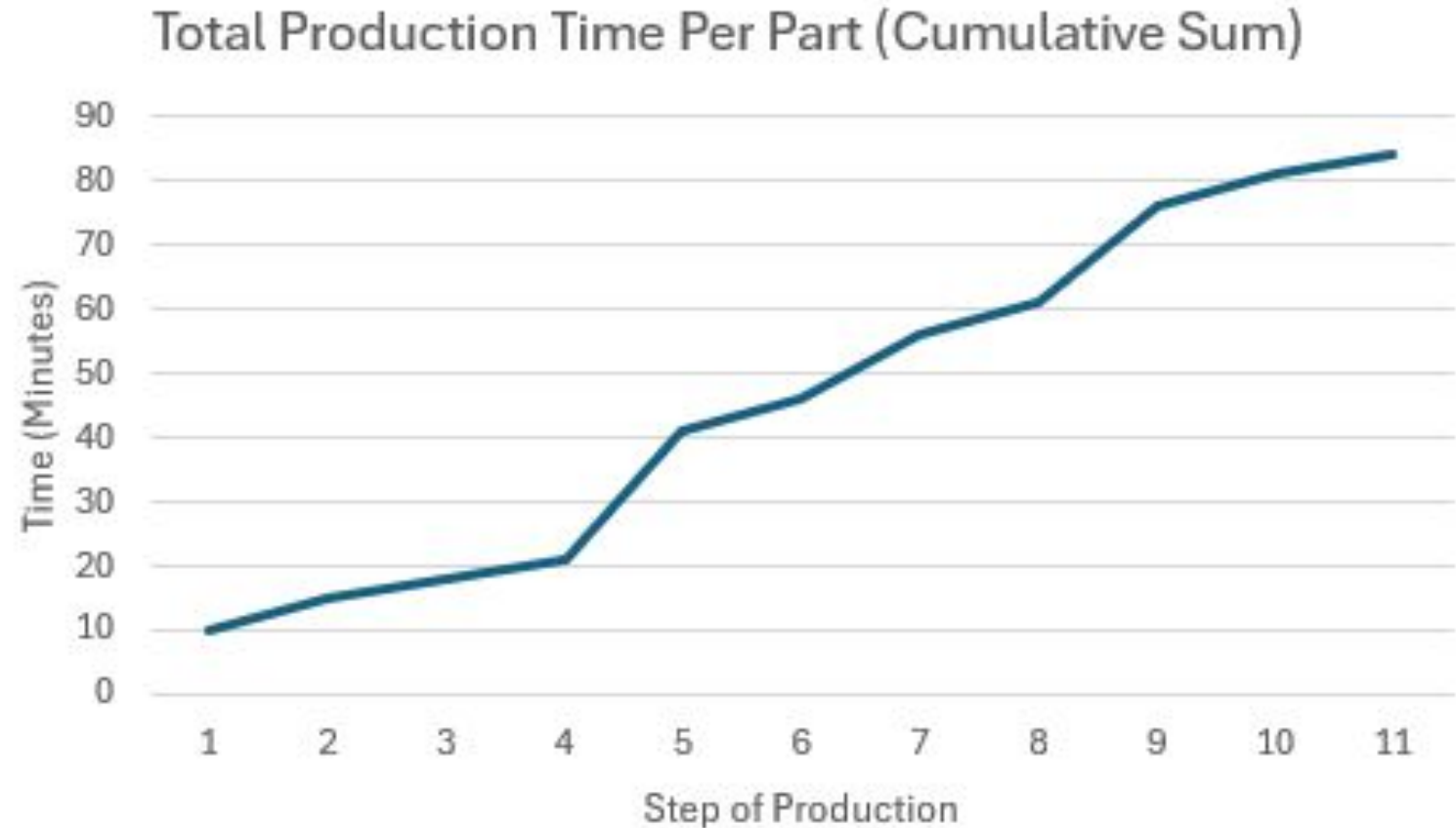


Pre-Process

1. Billet to Blank
2. Machine Setup
3. CAM G-Code

4. Fixturing
5. Milling
6. De-Fixturing

7. Remove Tubes
8. Deburr
9. Polishing
10. Cleaning
11. Inspecting

[illegible]



Sale Price

- \$350 Dollars
 - General Market Price
 - Chat GPT gave similar estimated price
 - About 10x the cost of material
 - Price Range \$250-\$600



The DF3 is entirely CNC screws on the bottom r densities to precisely ta balance each putter.

Machined right here in ti then assembled and han Creswell, Oregon.

\$559.00

4 interest-free installments, or fr
[Check your purchasing power](#)



PGM TUG049 novelty cnc milled putter 3 models transformable golf putter

\$499.00 USD

Home > Allison Slant Neck (Right and Left Hand)



ALLISON SLANT NECK (RIGHT AND LEFT HAND)

\$279.00 ~~\$479.00~~

4 interest-free installments, or from **\$25.18/mo** with [shop Pay](#)
[Check your purchasing power](#)

Select Hand

Left Hand

Select Shaft

34in Black



Annual Revenue

Economy Analysis for 1 Year Forecast	
Sales Price	\$/Unit
CNC Custom Golf Putter	\$ 350.00
Cost of Goods Sold	\$/Unit
Materials	\$ 35.00
Direct Labor	\$ 51.20
CNC Machine	\$ 307.69
Tooling & Fixturing	\$ 3.60
Consumable	\$ 3.85
Building	\$ 15.38
Shipping	\$ 13.08
Energy	\$ 0.77
Gross Profit	\$/Unit
Profit	\$ (80.57)
Gross Profit Percentage	-23%
General Overhead Expenses	\$/Unit
Computers	\$ 3.85
Software	\$ 8.85
Non-direct Labor	\$ 25.60
Advertising	\$ 6.15
Insurance	\$ 3.85
Net Profit	\$/Unit
Profit	\$ (128.86)
Total Profit	\$/Year
Total Profit	\$ (167,524.00)

Economy Analysis for 3 Year Forecast	
Sales Price	\$/Unit
CNC Custom Golf Putter	\$ 350.00
Cost of Goods Sold	\$/Unit
Materials	\$ 35.00
Direct Labor	\$ 53.57
CNC Machine	\$ 102.56
Tooling & Fixturing	\$ 3.60
Consumable	\$ 3.85
Building	\$ 16.10
Shipping	\$ 13.08
Energy	\$ 0.77
Gross Profit	\$/Unit
Profit	\$ 121.47
Gross Profit Percentage	34.71%
General Overhead Expenses	\$/Unit
Computers	\$ 1.28
Software	\$ 8.85
Non-direct Labor	\$ 26.79
Advertising	\$ 6.44
Insurance	\$ 4.02
Net Profit	\$/Unit
Profit	\$ 74.09
Total Profit	\$/Year
Total Profit	\$ 96,317.07

Economy Analysis for 5 Year Forecast	
Sales Price	\$/Unit
CNC Custom Golf Putter	\$ 350.00
Cost of Goods Sold	\$/Unit
Materials	\$ 35.00
Direct Labor	\$ 55.28
CNC Machine	\$ 61.54
Tooling & Fixturing	\$ 3.60
Consumable	\$ 3.85
Building	\$ 16.61
Shipping	\$ 13.08
Energy	\$ 0.77
Gross Profit	\$/Unit
Profit	\$ 160.28
Gross Profit Percentage	46%
General Overhead Expenses	\$/Unit
Computers	\$ 0.77
Software	\$ 8.85
Non-direct Labor	\$ 27.64
Advertising	\$ 6.64
Insurance	\$ 4.15
Net Profit	\$/Unit
Profit	\$ 112.23
Total Profit	\$/Year
Total Profit	\$ 145,897.02

Sustainability Impacts



Oregon State University
College of Engineering

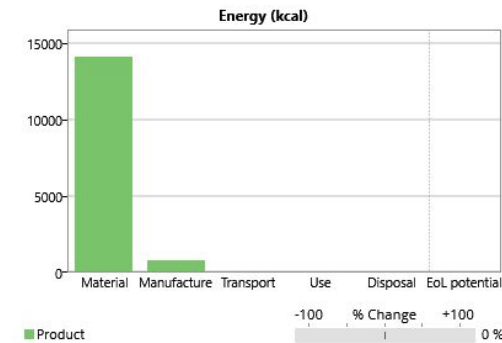
Society

- Pros
 - Job
 - Local Economy
 - Purpose
 - Happiness to Customer
- Cons
 - Energy & CO2 Emissions
 - Noise pollution

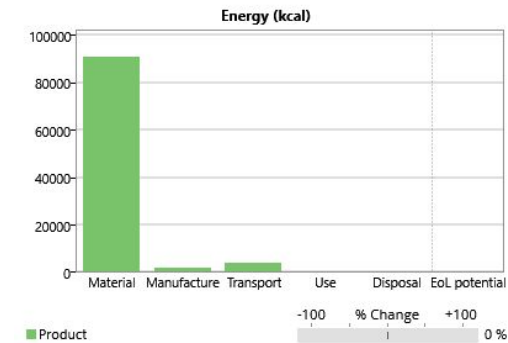
Environmental

- Expected
 - Aluminum
 - Cardboard
 - Cargo Boat
- Changed
 - Titanium
 - Polystyrene
 - Aircraft

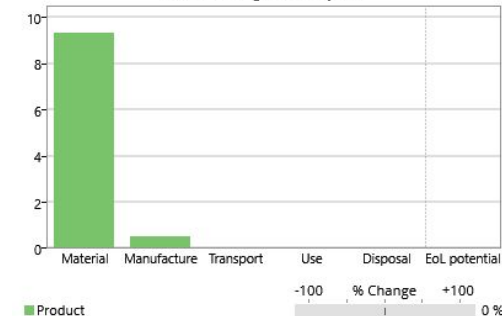
Expected Settings



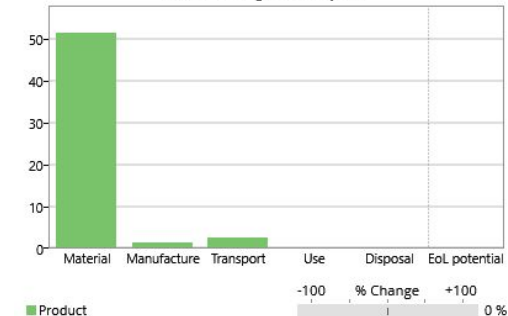
Changed Settings



Climate change (CO2-eq) (lb)



Climate change (CO2-eq) (lb)





Conclusion Discussion

- Automatization and Palletization
 - Currently 92% machine downtime
- Market For Product
 - Competitive Market
 - Low Purchase Frequency
- Hard to make profitable business from ground up